

First Section.
The Elliptic Integrals.

§ 7. Elliptic space curves of fourth order.

One can also represent algebraic functions of one variable by equations between several variables, if the number of equations is increased correspondingly. For example, take three variables x, y, z and let two equations

$$\varphi(x, y, z) = 0, \quad \psi(x, y, z) = 0 \quad (1)$$

hold between them. Then from these two equations one can eliminate, for example, x and obtains an equation between y and z by which y is defined as an algebraic function of z . Then, if certain exceptional cases are excluded, x and every rational function of x, y, z can be represented rationally by y and z . The integrals of the form

$$\int F(x, y, z) dz, \quad (2)$$

in which F denotes a rational function, then belong to the algebraic configuration represented by (1). If x, y, z are taken as Cartesian coordinates in space, then (1) represents a space curve as the intersection of two surfaces $\varphi = 0, \psi = 0$, and the integrals (2) belong to this space curve. The curve is again called elliptic when these integrals are elliptic.

We shall apply these considerations to the space curves of fourth order and first species, that is, to the curves of fourth order which can be represented as the complete intersection of two surfaces of the second degree.

We again introduce homogeneous coordinates x_1, x_2, x_3, x_4 and assume the equations of two given surfaces of the second degree in the form

$$\varphi(x_1, x_2, x_3, x_4) = \sum a_{ik} x_i x_k, \quad \psi(x_1, x_2, x_3, x_4) = \sum b_{ik} x_i x_k. \quad (3)$$

The two surfaces determine a pencil of surfaces of the second order $\xi\varphi = \eta\psi$. All surfaces of the pencil cut one another in a space curve of fourth order, which we call the base curve. One obtains the same curve and the same pencil if the functions φ and ψ are replaced by φ', ψ' , which are derived from φ, ψ by means of the linear substitution $\begin{pmatrix} m & n \\ p & q \end{pmatrix}$, that is, by

$$\varphi' = m\varphi + n\psi, \quad \psi' = p\varphi + q\psi, \quad (4)$$

whose determinant $r = mq - np$ is different from zero. This gives the identity

$$\xi\varphi + \eta\psi = \xi'\varphi' + \eta'\psi', \quad (5)$$

where

$$\xi = m\xi' + p\eta', \quad \eta = n\xi' + q\eta' \quad (6)$$

represents a linear transformation of the variables ξ, η . Besides this binary substitution there also comes into consideration a linear quaternary substitution of the variables x_1, x_2, x_3, x_4 (coordinate transformation), whose determinant we may assume to be equal to 1.

Every quadratic form has, with respect to these latter transformations, an invariant, namely the Hessian determinant (Vol. I, §§ 62, 63, 66), and we therefore obtain as invariant of the form (5)

$$f(\xi, \eta) = \sum \pm (\xi a_{11} + \eta b_{11})(\xi a_{22} + \eta b_{22})(\xi a_{33} + \eta b_{33})(\xi a_{44} + \eta b_{44}). \quad (7)$$

This is a binary biquadratic form in the variables ξ, η , which, when developed, we write as

$$f(\xi, \eta) = a_0\xi^4 + a_1\xi^3\eta + a_2\xi^2\eta^2 + a_3\xi\eta^3 + a_4\eta^4. \quad (8)$$

The coefficients of this biquadratic form, a_0, a_1, a_2, a_3, a_4 , do not change under a coordinate transformation. They are therefore called simultaneous invariants of the pair of forms φ, ψ (a_0 and a_4 are the determinants of φ and of ψ).

The a_i change when ξ, η are subjected to a substitution (6). But if one forms the invariants of the form (8) (as in Algebra, Vol. I, § 70), one obtains functions of the coefficients which are also invariant under these substitutions, and which therefore do not belong to the individual forms φ, ψ , but to the whole pencil and consequently also to their intersection, the base curve. We call them invariants of the base curve. With respect to the system of forms φ, ψ they are also called combinants. Thus the base curve has two invariants:

$$A = a_2^2 - 3a_1a_3 + 12a_0a_4, \quad B = 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3, \quad (9)$$

from which one derives the discriminant D by the formula

$$27D = 4A^3 - B^2. \quad (10)$$

With respect to the coefficients of φ and ψ , the invariants A, B, D are of degrees 8, 12, 24.

The system of forms φ, ψ has two simultaneous quadratic covariants, which are derived in the same way as the covariant C (in Algebra I, § 65). The first of them is, if we put $\psi_i = \frac{1}{2} \frac{\partial \psi}{\partial x_i}$,

$$\Phi = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} & \psi_1 \\ a_{21} & a_{22} & a_{23} & a_{24} & \psi_2 \\ a_{31} & a_{32} & a_{33} & a_{34} & \psi_3 \\ a_{41} & a_{42} & a_{43} & a_{44} & \psi_4 \\ \psi_1 & \psi_2 & \psi_3 & \psi_4 & 0 \end{vmatrix}, \quad (11)$$

and the second, Ψ , is obtained from this by interchanging a_{ik} with b_{ik} and ψ with φ .

The equation $\Phi = 0$ expresses a surface of the second degree (not belonging to the pencil), which is the geometric locus of the points x whose polar planes, with respect to ψ , touch the surface φ .

We now assume that the discriminant D is different from zero. Then the two functions φ, ψ can be transformed simultaneously into the sum of four squares:

$$\varphi = \alpha_1 y_1^2 + \alpha_2 y_2^2 + \alpha_3 y_3^2 + \alpha_4 y_4^2, \quad \psi = \beta_1 y_1^2 + \beta_2 y_2^2 + \beta_3 y_3^2 + \beta_4 y_4^2, \quad (12)$$

where the y_i are linear functions of the x_i . This is the canonical form of the pair of functions. Then

$$f(\xi, \eta) = (\xi\alpha_1 + \eta\beta_1)(\xi\alpha_2 + \eta\beta_2)(\xi\alpha_3 + \eta\beta_3)(\xi\alpha_4 + \eta\beta_4), \quad (13)$$

and the functions Φ, Ψ receive the canonical form

$$\Phi = \sum_{i=1}^4 \beta_i^2 \prod_{j \neq i} \alpha_j y_i^2, \quad \Psi = \sum_{i=1}^4 \alpha_i^2 \prod_{j \neq i} \beta_j y_i^2. \quad (14)$$

The determinant of the system (12), (14), considered as linear equations for $y_1^2, y_2^2, y_3^2, y_4^2$, is

$$\begin{vmatrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \\ \beta_1 & \beta_2 & \beta_3 & \beta_4 \\ \beta_1^2 \alpha_2 \alpha_3 \alpha_4 & \beta_2^2 \alpha_1 \alpha_3 \alpha_4 & \beta_3^2 \alpha_1 \alpha_2 \alpha_4 & \beta_4^2 \alpha_1 \alpha_2 \alpha_3 \\ \alpha_1^2 \beta_2 \beta_3 \beta_4 & \alpha_2^2 \beta_1 \beta_3 \beta_4 & \alpha_3^2 \beta_1 \beta_2 \beta_4 & \alpha_4^2 \beta_1 \beta_2 \beta_3 \end{vmatrix}. \quad (15)$$

It vanishes when $\alpha_i : \alpha_k = \beta_i : \beta_k$, because then the first two columns become proportional to one another, and likewise when $\alpha_i : \alpha_k = \beta_i : \beta_k$. Therefore (15), considered as an integral function of the α, β , is divisible by

$$\alpha_i \beta_k - \alpha_k \beta_i = (\alpha_i \beta_k), \quad (16)$$

and hence also by the product of all these factors

$$(\alpha_1 \beta_2)(\alpha_1 \beta_3)(\alpha_1 \beta_4)(\alpha_2 \beta_3)(\alpha_2 \beta_4)(\alpha_3 \beta_4) = \Delta. \quad (17)$$

Comparison of the degrees shows that the two functions (15), (17) can differ only by a numerical factor, and this factor is found from the assumption $\beta_1 = 0, \alpha_2 = 0, \alpha_4 = 0$ to be equal to 1. The determinant (15) is therefore equal to the product Δ , and the square of Δ is the discriminant D of the function $f(\xi, \eta)$, hence is equal to the function D , which we have assumed to be different from zero. Thus the $y_1^2, y_2^2, y_3^2, y_4^2$ can be represented linearly through $\varphi, \psi, \Phi, \Psi$.

The covariants Φ, Ψ change when φ and ψ are replaced by φ', ψ' according to (4). They therefore do not belong to the base curve, but to the pair of forms φ, ψ . But under this substitution Φ and Ψ , as linear functions of $y_1^2, y_2^2, y_3^2, y_4^2$, pass into linear functions of $\varphi, \psi, \Phi, \Psi$. These expressions are rather complicated. We need them here, however, only for the points of the base curve, that is, for $\varphi = 0, \psi = 0$, and under this assumption they become very simple. Namely

$$\begin{aligned} \Phi' &= \sum (p\alpha_1 + q\beta_1)^2 (m\alpha_2 + n\beta_2)(m\alpha_3 + n\beta_3)(m\alpha_4 + n\beta_4)y_1^2, \\ \Psi' &= \sum (m\alpha_1 + n\beta_1)^2 (p\alpha_2 + q\beta_2)(p\alpha_3 + q\beta_3)(p\alpha_4 + q\beta_4)y_1^2. \end{aligned} \quad (18)$$

and from this, by (12) and (14), one has to derive

$$\Phi' = M\Phi + N\Psi, \quad \Psi' = P\Phi + Q\Psi,$$

where the coefficients M, N, P, Q still have to be determined. We can carry this out by computation. Without computation the result is obtained as follows.

The M, N are integral functions of degree two in p, q and of degree three in m, n . If one now takes $p : q = m : n$, so that $r = mq - np = 0$, then Φ' and $\partial\Phi'/\partial p$ pass by (18) into linear combinations of φ and ψ , and thus vanish. It follows that M and N are divisible by r^2 , and the quotients are linear functions of m and n . But since in the two cases

$$\begin{pmatrix} m & n \\ p & q \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Φ' must pass into Φ and into Ψ , and since the same consideration can be applied to Ψ' , it follows that

$$\Phi' = r^2(m\Phi + n\Psi), \quad \Psi' = r^2(p\Phi + q\Psi). \quad (19)$$

Thus, apart from the factor r^2 , the functions Φ, Ψ are transformed cogrediently with φ, ψ . (For the strict justification of these conclusions see Vol. I, § 20.)

From these results we can in a simple way, always under the assumption $\varphi = 0, \psi = 0$, express the y_i^2 by Φ and Ψ , that is, solve the linear equations (12), (14). If we apply the formulae (19) to the substitution

$$\begin{pmatrix} m & n \\ p & q \end{pmatrix} = \begin{pmatrix} \beta_1 & -\alpha_1 \\ 0 & 1 \end{pmatrix},$$

then $r = \beta_1$, the function ψ' remains unchanged $= \psi$, and φ' is obtained from φ if

$$\alpha_1, \alpha_2, \alpha_3, \alpha_4$$

are replaced by

$$0, \quad (\alpha_2\beta_1), \quad (\alpha_3\beta_1), \quad (\alpha_4\beta_1).$$

Thus by (14)

$$\Phi' = \beta_1^2(\alpha_2\beta_1)(\alpha_3\beta_1)(\alpha_4\beta_1)y_1^2,$$

and consequently by (19)

$$\begin{aligned} (\alpha_2\beta_1)(\alpha_3\beta_1)(\alpha_4\beta_1)y_1^2 &= \alpha_1\Phi - \beta_1\Psi, \\ (\alpha_1\beta_2)(\alpha_3\beta_2)(\alpha_4\beta_2)y_2^2 &= \alpha_2\Phi - \beta_2\Psi, \\ (\alpha_1\beta_3)(\alpha_2\beta_3)(\alpha_4\beta_3)y_3^2 &= \alpha_3\Phi - \beta_3\Psi, \\ (\alpha_1\beta_4)(\alpha_2\beta_4)(\alpha_3\beta_4)y_4^2 &= \alpha_4\Phi - \beta_4\Psi. \end{aligned} \tag{20}$$

The product of the constant factors on the left is the discriminant D , and therefore by (13) one obtains

$$Dy_1^2y_2^2y_3^2y_4^2 = f(\Phi, -\Psi) = a_0\Phi^4 - a_1\Phi^3\Psi + a_2\Phi^2\Psi^2 - a_3\Phi\Psi^3 + a_4\Psi^4, \tag{21}$$

and the y_i^2 are therefore the linear factors of this biquadratic form.

There is still another simultaneous covariant of the form φ, ψ , namely the Jacobi functional determinant of the four functions $\varphi, \psi, \Phi, \Psi$. We denote it by

$$K = \begin{vmatrix} \varphi_1 & \varphi_2 & \varphi_3 & \varphi_4 \\ \psi_1 & \psi_2 & \psi_3 & \psi_4 \\ \Phi_1 & \Phi_2 & \Phi_3 & \Phi_4 \\ \Psi_1 & \Psi_2 & \Psi_3 & \Psi_4 \end{vmatrix}. \tag{22}$$

If it is formed for the canonical form, then by (12), (14), (15) one obtains

$$K = -\Delta y_1 y_2 y_3 y_4,$$

and since $\Delta^2 = D$,

$$K^2 = Dy_1^2y_2^2y_3^2y_4^2.$$

Therefore by (21) we have

$$K^2 = f(\Phi, -\Psi). \tag{23}$$

The biquadratic form $f(\Phi, -\Psi)$ has two covariants, which we denote as in Vol. I, § 70:

$$H = \frac{1}{3} \begin{vmatrix} \frac{\partial^2 f}{\partial \Phi^2} & -\frac{\partial^2 f}{\partial \Phi \partial \Psi} \\ \frac{\partial^2 f}{\partial \Psi \partial \Phi} & -\frac{\partial^2 f}{\partial \Psi^2} \end{vmatrix}, \quad T = \frac{1}{12} \begin{vmatrix} f'(\Phi) & f'(-\Psi) \\ H'(\Phi) & H'(-\Psi) \end{vmatrix}. \tag{24}$$

They are of the fourth and sixth degrees in Φ, Ψ , hence of the eighth and twelfth degrees in the x_i . They are covariants of the base curve (combinants of φ and ψ), and between them there is the relation

$$H^3 - 48AHf^2 - 64Bf^3 = -27T^2. \tag{25}$$

The most general integral (2) belonging to the base curve passes, by introduction of homogeneous variables $z = x_3 : x_4$, into the following form:

$$\int F'(x, y, z) \frac{x_3 dx_4 - x_4 dx_3}{x_4^2},$$

or, if one puts

$$F(x_1, x_2, x_3, x_4) = x_4^2 \frac{F'(x, y, z)}{\varphi_1\psi_2 - \varphi_2\psi_1},$$

into

$$\int F(x_1, x_2, x_3, x_4) \frac{x_3 dx_4 - x_4 dx_3}{\varphi_1 \psi_2 - \varphi_2 \psi_1},$$

where F is a homogeneous function of degree zero. The simplest case is $F = 1$, and we therefore consider the integral of the first kind

$$u = \int \frac{x_3 dx_4 - x_4 dx_3}{\varphi_1 \psi_2 - \varphi_2 \psi_1}. \quad (26)$$

Now, by the multiplication theorem for determinants, we form the product $K(x_3 dx_4 - x_4 dx_3)$, hence, since $\varphi = 0, \psi = 0$,

$$\begin{vmatrix} \varphi_1 & \varphi_2 & \varphi_3 & \varphi_4 \\ \psi_1 & \psi_2 & \psi_3 & \psi_4 \\ \Phi_1 & \Phi_2 & \Phi_3 & \Phi_4 \\ \Psi_1 & \Psi_2 & \Psi_3 & \Psi_4 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ x_1 & x_2 & x_3 & x_4 \\ dx_1 & dx_2 & dx_3 & dx_4 \end{vmatrix} = \begin{vmatrix} \varphi_1 & \varphi_2 & 0 & 0 \\ \psi_1 & \psi_2 & 0 & 0 \\ \Phi_1 & \Phi_2 & d\Phi & \Phi \\ \Psi_1 & \Psi_2 & d\Psi & \Psi \end{vmatrix} \\ = (\varphi_1 \psi_2 - \varphi_2 \psi_1)(\Phi d\Psi - \Psi d\Phi),$$

and consequently from (26) and (23) one obtains

$$u = \int \frac{\Phi d\Psi - \Psi d\Phi}{K} = \int \frac{\Phi d\Psi - \Psi d\Phi}{\sqrt{f(\Phi, -\Psi)}}. \quad (27)$$

If one now applies the transformation of § 5 to this, replacing x, y by $\Phi, -\Psi$, one finds

$$u = \sqrt{3} \int \frac{f dH - H df}{\sqrt{-(H^3 - 48Af^2H - 64Bf^3)} f},$$

and if in § 5, (17), one puts $\mu = 3$, that is,

$$z = -\frac{3H}{4f},$$

then

$$u = 3 \int \frac{dz}{\sqrt{4z^3 - g_2z - g_3}}, \quad g_2 = 12 \cdot 9A, \quad g_3 = -4 \cdot 27B.$$

§ 8. Jacobi's transformation principle.

Jacobi's theory of transformation poses, in general, the problem of reducing an elliptic differential to another elliptic differential by an algebraic substitution. In setting out here the foundations of this theory, we use the representation of the elliptic differential in homogeneous variables [§ 1, (4)]:

$$\frac{x dy - y dx}{\sqrt{f(x, y)}}, \quad (1)$$

and substitute for x, y two relatively prime integral homogeneous functions of equal, but arbitrary, degree n in two new variables ξ, η :

$$x = U(\xi, \eta), \quad y = V(\xi, \eta). \quad (2)$$

From the equations

$$\begin{aligned} nU &= \xi \frac{\partial U}{\partial \xi} + \eta \frac{\partial U}{\partial \eta}, & dU &= d\xi \frac{\partial U}{\partial \xi} + d\eta \frac{\partial U}{\partial \eta}, \\ nV &= \xi \frac{\partial V}{\partial \xi} + \eta \frac{\partial V}{\partial \eta}, & dV &= d\xi \frac{\partial V}{\partial \xi} + d\eta \frac{\partial V}{\partial \eta}, \end{aligned}$$

there follows

$$U dV - V dU = H(\xi d\eta - \eta d\xi), \quad (3)$$

where H denotes the functional determinant

$$H = \frac{1}{n} \left(\frac{\partial U}{\partial \xi} \frac{\partial V}{\partial \eta} - \frac{\partial V}{\partial \xi} \frac{\partial U}{\partial \eta} \right). \quad (4)$$

Accordingly the elliptic differential (1) passes into

$$H \frac{\xi d\eta - \eta d\xi}{\sqrt{f(U, V)}}. \quad (5)$$

In order that this differential should again acquire the form of an elliptic differential, it is necessary that from the function $f(U, V)$, whose degree is $4n$, a square factor T^2 of degree $4n - 4$ be separated, hence, if $\varphi(\xi, \eta)$ denotes a function of the fourth degree, that

$$f(U, V) = T^2 \varphi(\xi, \eta) \quad (6)$$

become true, whereby, if

$$\frac{T}{H} = M \quad (7)$$

is put, the differential (5) passes into

$$\frac{1}{M} \frac{\xi d\eta - \eta d\xi}{\sqrt{\varphi(\xi, \eta)}}. \quad (8)$$

It can now be proved that as soon as the condition (6) is fulfilled, H is divisible by T , and therefore, since the degree of both functions is the same, M becomes a constant. This constant is called the multiplier of the transformation.

Indeed, observe that, since $f(x, y)$ contains no quadratic factor, the two functions

$$\frac{\partial f}{\partial x}, \quad \frac{\partial f}{\partial y}$$

have no common divisor, and that consequently, because U and V are relatively prime, also

$$\frac{\partial f}{\partial U}, \quad \frac{\partial f}{\partial V}$$

have no common divisor (with respect to ξ, η). Thus two functions α, β of ξ, η can be chosen in such a way that

$$\alpha \frac{\partial f}{\partial U} + \beta \frac{\partial f}{\partial V}$$

is relatively prime to an arbitrarily given function T . For example, one can assume α relatively prime to T , and β divisible by the divisors of T which do not occur in $\frac{\partial f}{\partial U}$, but not divisible by the common divisors of T and $\frac{\partial f}{\partial U}$. If we now differentiate the equation (6), which we suppose fulfilled, with respect to ξ, η , we get

$$\frac{\partial f}{\partial U} \frac{\partial U}{\partial \xi} + \frac{\partial f}{\partial V} \frac{\partial V}{\partial \xi} = TX,$$

$$\frac{\partial f}{\partial U} \frac{\partial U}{\partial \eta} + \frac{\partial f}{\partial V} \frac{\partial V}{\partial \eta} = TY,$$

and from this, by solving with respect to $\frac{\partial f}{\partial U}, \frac{\partial f}{\partial V}$:

$$H \left(\alpha \frac{\partial f}{\partial U} + \beta \frac{\partial f}{\partial V} \right) = TZ,$$

where X, Y, Z are integral homogeneous functions of ξ, η . From the last equation, however, one concludes that H must be divisible by T .

Accordingly our whole problem is contained in the equation (6), which says nothing other than that the function $f(U, V)$ of degree $4n$ is to contain $2n - 2$ quadratic factors. To satisfy the $2n - 2$ conditions resulting from this, one has at one's disposal the $2n + 2$ coefficients contained in U, V , so that four of them remain undetermined. This was to be foreseen, since arbitrary homogeneous linear functions of ξ, η can be introduced for ξ, η . These four remaining constants can be used in order to bring the differential (8) into a normal form. But since the equations of condition for the coefficients of U, V are not linear, there are several transformations for a given degree of transformation.

We shall later encounter this transformation problem again from an entirely different side and shall have to enter much more deeply into it. Therefore the details of the algebraic problem are not pursued further here; instead we shall illustrate what has been said by two examples.

§ 9. The transformation of the second degree.

We put, in order to obtain the elliptic differentials in normal form,

$$f(x, y) = xy(x - y)(x - \lambda^2 y), \quad \varphi(\xi, \eta) = \xi\eta(\xi - \eta)(\xi - \kappa^2 \eta),$$

and let U, V be of the second degree.

The equation (6), § 8, now becomes, since T is of the second degree,

$$UV(U - V)(U - \lambda^2 V) = (a\xi + b\eta)^2(a'\xi + b'\eta)^2\xi\eta(\xi - \eta)(\xi - \kappa^2 \eta), \quad (1)$$

and two of the four factors of the second degree on the left side of this equation must be squares. We shall further restrict the large number of possibilities lying here by supposing that η and y are to vanish simultaneously, so that V is divisible by η . This gives (apart from a constant factor) the following three possibilities for V :

$$1. V = \xi\eta, \quad 2. V = \eta(\xi - \eta), \quad 3. V = \eta(\xi - \kappa^2 \eta).$$

Each of these three cases again comprises three subcases, according as any two of the three remaining factors $U, U - V, U - \lambda^2 V$ are assumed to be squares. Of these latter three cases, two pass into one another by interchanging V, λ^2 with $V : \lambda^2, 1 : \lambda^2$.

We shall carry out completely two transformations among these which are especially important for what follows.

1. The Gaussian transformation.

$$V = \xi\eta, \quad U = (a\xi + b\eta)^2.$$

If now $U - \lambda^2 V$ is also to be a square, one must put

$$\lambda^2 = 4ab,$$

and then

$$U - \lambda^2 V = (a\xi - b\eta)^2.$$

Since $U - V$ must then be divisible by $\xi - \eta$ and by $\xi - \kappa^2 \eta$, one obtains, by putting $\xi = \eta$, $\xi = \kappa^2 \eta$ and using $U = V$, the conditions

$$a + b = \pm 1, \quad a\kappa^2 + b = \pm \kappa.$$

If the upper signs are taken, it follows that

$$\begin{aligned} a &= \frac{1}{1 + \varkappa}, & b &= \frac{\varkappa}{1 + \varkappa}, & \lambda &= \frac{2\sqrt{\varkappa}}{1 + \varkappa}, \\ V &= \xi\eta, & U &= \left(\frac{\xi + \varkappa\eta}{1 + \varkappa}\right)^2, & U - \lambda^2 V &= \left(\frac{\xi - \varkappa\eta}{1 + \varkappa}\right)^2, \\ U - V &= \frac{(\xi - \eta)(\xi - \varkappa^2\eta)}{(1 + \varkappa)^2}, \\ T &= \frac{\xi^2 - \varkappa^2\eta^2}{(1 + \varkappa)^3}, & H &= \frac{\xi^2 - \varkappa^2\eta^2}{(1 + \varkappa)^2}, & M &= \frac{1}{1 + \varkappa}. \end{aligned}$$

Putting again

$$\frac{y}{x} = z, \quad \frac{\eta}{\xi} = \zeta,$$

we therefore have the result that the substitution

$$z = \frac{(1 + \varkappa)^2 \zeta}{(1 + \varkappa \zeta)^2}, \quad \lambda^2 = \frac{4\varkappa}{(1 + \varkappa)^2} \quad (2)$$

effects the transformation

$$\frac{dz}{\sqrt{z(1-z)(1-\lambda^2 z)}} = \frac{(1 + \varkappa)d\zeta}{\sqrt{\zeta(1-\zeta)(1-\varkappa^2 \zeta)}}. \quad (3)$$

2. The Landen transformation.

$$V = a\eta(\xi - \eta), \quad U = \xi(\xi - \varkappa^2\eta).$$

The conditions that $U - V$, $U - \lambda^2 V$ be squares are

$$4a = (\varkappa^2 + a)^2, \quad 4a\lambda^2 = (\varkappa^2 + a\lambda^2)^2,$$

whence

$$\lambda(\varkappa^2 + a) = \varkappa^2 + a\lambda^2,$$

or, after division by $\lambda - 1$,

$$a\lambda = \varkappa^2.$$

Substitution of this into one of the above equations gives

$$4\lambda = \varkappa^2(1 + \lambda)^2,$$

and by solving this quadratic equation, if $\varkappa' = \sqrt{1 - \varkappa^2}$ is put,

$$\begin{aligned} \lambda &= \frac{1 - \varkappa'}{1 + \varkappa'}, & a &= (1 + \varkappa')^2, \\ U - V &= \{\xi - (1 + \varkappa')\eta\}^2, & U - \lambda^2 V &= \{\xi - (1 - \varkappa')\eta\}^2, \\ H &= (1 + \varkappa')^2 \{\xi - (1 + \varkappa')\eta\} \{\xi - (1 - \varkappa')\eta\}, \\ T &= (1 + \varkappa') \{(\xi - \eta)^2 - \varkappa'^2 \eta^2\}, & M &= \frac{1}{1 + \varkappa'}. \end{aligned}$$

Thus by the substitution

$$z = \frac{(1 + \varkappa')^2 \zeta(1 - \zeta)}{1 - \varkappa'^2 \zeta}, \quad \lambda = \frac{1 - \varkappa'}{1 + \varkappa'}, \quad (4)$$

one obtains the transformation

$$\frac{dz}{\sqrt{z(1-z)(1-\lambda^2 z)}} = \frac{(1+\kappa') d\zeta}{\sqrt{\zeta(1-\zeta)(1-\kappa'^2 \zeta)}}. \quad (5)$$

If one again applies the Gaussian transformation to the left side of this equation, replacing in formulae (2), (3) ξ, λ by z, λ and replacing z by a variable η , then, as follows from (2) and (4), λ in (3) is to be replaced by κ , and by combining (5) and (3) one finds

$$\frac{d\eta}{\sqrt{\eta(1-\eta)(1-\kappa^2 \eta)}} = \frac{2 d\zeta}{\sqrt{\zeta(1-\zeta)(1-\kappa^2 \zeta)}}, \quad (6)$$

and the connection of (2) and (4) gives the following relation between the variables η and ζ :

$$\eta = \frac{4\zeta(1-\zeta)(1-\kappa^2 \zeta)}{(1-\kappa^2 \zeta^2)^2}. \quad (7)$$

The combination of the two transformations of the second degree therefore gives multiplication by 2.

§ 10. The transformation of the third degree.

As a second example we consider the transformation of the third degree. The equation

$$UV(U-V)(U-\lambda^2 V) = T^2 \xi \eta (\xi - \eta) (\xi - \kappa^2 \eta) \quad (1)$$

requires that each of the four factors of the third degree, $U, V, U-V, U-\lambda^2 V$, be divisible by one of the linear factors $\xi, \eta, \xi - \eta, \xi - \kappa^2 \eta$, and that the quotient be a square. We therefore put

$$U = \xi(a\xi + b\eta)^2, \quad V = \eta(a'\xi + b'\eta)^2, \quad (2)$$

and further require that

$$\frac{U-V}{\xi-\eta}, \quad \frac{U-\lambda^2 V}{\xi-\kappa^2 \eta} \quad (3)$$

become the squares of linear functions.

Of the two latter requirements, one follows from the other if U, V are arranged so that, apart from constant factors, U and V pass into one another by the interchange of ξ, η with $\kappa^2 \eta, \xi$. Under this interchange, however,

$$\xi(a\xi + b\eta)^2, \quad \eta(a'\xi + b'\eta)^2$$

passes into

$$\kappa^2 \eta(b\xi + a\kappa^2 \eta)^2, \quad \xi(b'\xi + a'\kappa^2 \eta)^2,$$

and our requirement is fulfilled if

$$\kappa^2 = \frac{bb'}{aa'} \quad (4)$$

holds. If then

$$\lambda^2 = \kappa^2 \left(\frac{ab}{a'b'} \right)^2 \quad (5)$$

is true, this interchange carries

$$\frac{U-V}{\xi-\eta} \quad \text{into} \quad \frac{b'^2 U - \lambda^2 V}{a^2 \xi - \kappa^2 \eta}. \quad (6)$$

After this has been fixed, there remains only the condition that the first of the two quantities (3) be the square of a linear function, which we denote by $(\alpha^2\xi - \beta^2\eta)$, that is,

$$U - V = (\xi - \eta)(\alpha^2\xi - \beta^2\eta)^2 \quad (7)$$

or

$$U - \xi(\alpha^2\xi - \beta^2\eta)^2 = V - \eta(\alpha^2\xi - \beta^2\eta)^2.$$

Since U is divisible by ξ and V by η , this function must be divisible by $\xi\eta$; if we put it equal to $\xi\eta(m\xi + n\eta)$, then from (2) we obtain

$$(a\xi + b\eta)^2 = (\alpha^2\xi - \beta^2\eta)^2 + \eta(m\xi + n\eta),$$

$$(a'\xi + b'\eta)^2 = (\alpha^2\xi - \beta^2\eta)^2 + \xi(m\xi + n\eta).$$

Comparison of coefficients gives

$$a = \alpha^2, \quad b = -\beta^2 + \frac{m}{2\alpha^2}, \quad b^2 = \beta^4 + n,$$

$$b' = \beta^2, \quad a' = -\alpha^2 + \frac{n}{2\beta^2}, \quad a'^2 = \alpha^4 + m,$$

and from the two last equations of each row,

$$m^2 = 4\alpha^2(n\alpha^2 + m\beta^2), \quad n^2 = 4\beta^2(n\alpha^2 + m\beta^2).$$

If therefore one puts

$$m = h\alpha, \quad n = h\beta,$$

then

$$h = 4\alpha\beta(\alpha + \beta).$$

Accordingly a, b, a', b' can be expressed by the two quantities α, β in the form

$$a = \alpha^2, \quad b = \beta^2 + 2\alpha\beta, \quad a' = \alpha^2 + 2\alpha\beta, \quad b' = \beta^2, \quad (8)$$

and then from (4), (5)

$$\varkappa^2 = \frac{\beta^3}{\alpha^3} \frac{\beta + 2\alpha}{\alpha + 2\beta}, \quad \lambda^2 = \frac{\alpha}{\beta} \frac{\beta + 2\alpha}{\alpha + 2\beta}, \quad (9)$$

or, by multiplying and dividing these two equations,

$$\sqrt{\lambda\varkappa} = \frac{\beta}{\alpha} \frac{\beta + 2\alpha}{\alpha + 2\beta}, \quad \frac{\varkappa^3}{\lambda} = \frac{\beta^4}{\alpha^4}. \quad (10)$$

By eliminating $\beta : \alpha$ one obtains an equation between $\varkappa, \lambda$, which is called the modular equation, and whose degree gives the number of distinct transformations of the third degree. It assumes the simplest form if one puts

$$\sqrt[4]{\varkappa} = u, \quad \sqrt[4]{\lambda} = v. \quad (11)$$

Then equations (10) become

$$u^2v^2 = \frac{\beta}{\alpha} \frac{\beta + 2\alpha}{\alpha + 2\beta}, \quad \frac{u^3}{v} = \frac{\beta}{\alpha},$$

and therefore, by substituting the value of $\beta : \alpha$ into the first equation and removing the factor u^2 ,

$$v^4 - u^4 + 2u^3v^3 - 2uv = 0, \quad (12)$$

an equation of the fourth degree.

Furthermore one obtains [without lengthy calculation, by comparing the coefficients of the highest power of ξ in (1)]

$$T = \frac{a}{b'}(a\xi + b\eta)(a'\xi + b'\eta)(\alpha^2\xi - \beta^2\eta)(\beta^2\xi - \kappa^2\alpha^2\eta),$$

$$H = \frac{1}{3} \left(\frac{\partial U}{\partial \xi} \frac{\partial V}{\partial \eta} - \frac{\partial V}{\partial \xi} \frac{\partial U}{\partial \eta} \right) = \frac{a'}{a} T,$$

whence one easily finds by (8)

$$M = \frac{a}{a'} = \frac{\alpha}{\alpha + 2\beta} = \frac{v}{v + 2u^3}. \quad (13)$$

If the expression derived from this,

$$v = \frac{2u^3 M}{1 - M},$$

is substituted into equation (12), one obtains for M an equation of the fourth degree, the multiplier equation, which can replace the modular equation. But one obtains this equation more simply in the following way. By (13)

$$\frac{1}{M} - 1 = 2\frac{\beta}{\alpha}, \quad \frac{1}{M} + 3 = \frac{2(\beta + 2\alpha)}{\alpha},$$

and hence by (9)

$$\left(\frac{1}{M} - 1 \right)^3 \left(\frac{1}{M} + 3 \right) = 16\kappa^2 \frac{\alpha + 2\beta}{\alpha},$$

and consequently by (13)

$$\frac{16\kappa^2}{M} = \left(\frac{1}{M} - 1 \right)^3 \left(\frac{1}{M} + 3 \right),$$

or, ordered,

$$\frac{1}{M^4} - \frac{6}{M^2} + \frac{8(1 - 2\kappa^2)}{M} - 3 = 0. \quad (14)$$

If we therefore express our formulae by u, v , the result of this consideration is the following. The substitution

$$z = \frac{\zeta[(v^2 + 2vu^3) + u^6\zeta]^2}{[v^2 + (u^6 + 2vu^3)\zeta]^2}$$

effects the transformation

$$\frac{dz}{\sqrt{z(1-z)(1-v^8z)}} = \frac{v + 2u^3}{v} \frac{d\zeta}{\sqrt{\zeta(1-\zeta)(1-u^8\zeta)}},$$

provided the modular equation (12) holds between u, v .

For a given u , the modular equation gives four values of v , hence four distinct transformations of the third degree.

§ 11. The three kinds of elliptic integrals.

Let now

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 \quad (1)$$

be an integral function of the third or fourth degree, and

$$\Omega(x) = \int \frac{\Phi(x) dx}{\sqrt{f(x)}} = \int R(x) dx, \quad (2)$$

where, for abbreviation,

$$R(x) = \frac{\Phi(x)}{\sqrt{f(x)}} \quad (3)$$

has been put, an elliptic integral. Here x denotes an unrestrictedly variable (also complex) quantity, and $\sqrt{f(x)}$ has, for each value of x for which $f(x)$ does not vanish, two opposite values. By a point we understand not a value of x alone, but an associated pair of values of x and $\sqrt{f(x)}$, hence a value of x with a definite sign of $\sqrt{f(x)}$. Thus if x_0 is any value of x , there are two points x_0 if $f(x_0)$ is different from zero, but only one if $f(x_0) = 0$. For $x = \infty$ we determine the points according to the sign of $\sqrt{f(x)} : x^2$, and therefore we have two points $x = \infty$ if $f(x)$ is of the fourth degree, and only one if $f(x)$ is of the third degree.

By known theorems of function theory, if $f(x_0)$ is different from zero, there is in a certain region a convergent expansion according to Taylor's theorem:

$$R(x) = A_m(x - x_0)^m + A_{m+1}(x - x_0)^{m+1} + A_{m+2}(x - x_0)^{m+2} + \dots, \quad (4)$$

where m is a positive or negative integer or also zero, while $A_m, A_{m+1}, A_{m+2}, \dots$ denote constants, of which A_m does not vanish. This is an expansion in ascending powers. If $f(x)$ is of the fourth (not of the third) degree, there is added the expansion in descending powers

$$R(x) = C_m x^{-m-2} + C_{m+1} x^{-m-3} + C_{m+2} x^{-m-4} + \dots, \quad (5)$$

of which we say that it holds in a neighborhood of an infinitely distant point.

But if $f(x_0)$ vanishes, then the expansion of $R(x)$ receives the form

$$R(x) = A_m(x - x_0)^{m-\frac{1}{2}} + A_{m+1}(x - x_0)^{m-\frac{1}{2}+1} + A_{m+2}(x - x_0)^{m-\frac{1}{2}+2} + \dots, \quad (6)$$

and if $f(x)$ is of the third degree, then in a neighborhood of the infinitely distant point the expansion

$$R(x) = C_m x^{-m-\frac{3}{2}} + C_{m+1} x^{-m-\frac{5}{2}} + C_{m+2} x^{-m-\frac{7}{2}} + \dots \quad (7)$$

holds.

From the developments (4) to (7) we can derive the developments of Ω , an additive constant remaining undetermined.

The development of Ω is then likewise a power series, to which, if m is negative, in the cases (4) or (5) there is still added a term $A_{-1} \log(x - x_0)$, $C_{-1} \log x$.

We now call x_0 a point of the first kind of Ω if m in these developments is not negative. Then Ω is finite at the point x_0 (also in the case $x_0 = \infty$).

The point x_0 is called a point of the second kind of Ω if in the developments (4), (5) m is negative and $A_{-1} = 0$, $C_{-1} = 0$, and in the case of the developments (6), (7), with negative m .

Finally x_0 is called a point of the third kind if, in the developments (4) or (5), A_{-1} or C_{-1} is different from zero, that is, if a logarithmic term occurs in the development of Ω .

The zeros of $f(x)$ cannot be of the third kind for the integral (2); but they can become of the third kind if we consider the elliptic integral in the general form

$$\int \Phi(x, \sqrt{f(x)}) dx.$$

The integral Ω is called an integral of the first kind if it has only points of the first kind. Such an integral then has a finite value for all finite and infinite values of x .

Ω is called an integral of the second kind if it has only points of the first and second kinds, and Ω is called an integral of the third kind if, besides points of the first and second kinds, it also has points of the third kind.

§ 12. Representation of the elliptic integrals by the simplest fundamental integrals.

In order to represent the integral $\Omega(x)$ by integrals of the simplest possible type, we decompose the rational function $\Phi(x)$ into an integral function and a sum of partial fractions; that is, we represent $\Phi(x)$ as a sum of expressions of the form

$$x^n, \quad \frac{1}{(x - \alpha)^n}$$

with constant coefficients, where n denotes a positive integer (also zero) and α denotes a constant value for which $\Phi(x)$ becomes infinite.

If we then put, for positive and for negative or vanishing n ,

$$S_n(\alpha) = \int \frac{(x - \alpha)^n dx}{\sqrt{f(x)}}, \quad (1)$$

then $\Omega(x)$ becomes a sum of the form

$$\Omega(x) = \sum M S_n(\alpha),$$

where the sum may extend over several different values of n and of α , and the M are constants.

The functions S_n can be reduced, through the mediation of algebraic functions, to a small number. To this end we form

$$d\left((x - \alpha)^n \sqrt{f(x)}\right) = \frac{n(x - \alpha)^{n-1} f(x) + (x - \alpha)^{n-\frac{1}{2}} f'(x)}{\sqrt{f(x)}} dx,$$

and if in this we put

$$f(x) = f(\alpha) + (x - \alpha)f'(\alpha) + \frac{(x - \alpha)^2}{2}f''(\alpha) + \frac{(x - \alpha)^3}{6}f'''(\alpha) + \frac{(x - \alpha)^4}{24}f''''(\alpha),$$

$$f'(x) = f'(\alpha) + (x - \alpha)f''(\alpha) + \frac{(x - \alpha)^2}{2}f'''(\alpha) + \frac{(x - \alpha)^3}{6}f''''(\alpha),$$

then it follows that

$$\begin{aligned} (x - \alpha)^n \sqrt{f(x)} &= n f(\alpha) S_{n-1} + \left(n + \frac{1}{2}\right) f'(\alpha) S_n \\ &\quad + \frac{n+1}{2} f''(\alpha) S_{n+1} + \left(\frac{n}{6} + \frac{1}{4}\right) f'''(\alpha) S_{n+2} \\ &\quad + \left(\frac{n}{24} + \frac{1}{12}\right) f''''(\alpha) S_{n+3}. \end{aligned} \quad (2)$$

We distinguish four cases.

1. $f'''(\alpha)$ and $f(\alpha)$ are not $= 0$ [that is, $f(x)$ is of the fourth degree and α is no root of $f(x)$]. In this case, by formula (2) for $n = 0$, S_3 can be expressed through S_0, S_1, S_2 , and therefore S_n for every positive n through S_0, S_1, S_2 . But if one puts $n = -1, -2, \dots$, then one obtains $S_{-2}, S_{-3}, \dots$ expressed through S_{-1}, S_0, S_1, S_2 . Hence in this case there remain the four fundamental integrals

$$S_{-1}, \quad S_0, \quad S_1, \quad S_2.$$

2. $f'''(\alpha)$ is not $= 0$, $f(\alpha) = 0$. In this case S_{-1} can still be expressed through S_0, S_1, S_2 , and the fundamental integrals are

$$S_0, \quad S_1, \quad S_2.$$

3. $f'''(\alpha) = 0$, $f(\alpha) \neq 0$. Here the fundamental integrals are

$$S_{-1}, \quad S_0, \quad S_1.$$

4. $f'''(\alpha) = 0$, $f(\alpha) = 0$. Here the fundamental integrals are

$$S_0, \quad S_1.$$

The result is therefore:

I. If $f(x)$ is of the fourth degree ($a_0 = 1$), then all integrals $\Omega(x)$ can be expressed by integrals of the form

$$S_0 = \int \frac{dx}{\sqrt{f(x)}}, \quad S_1 = \int \frac{x dx}{\sqrt{f(x)}}, \quad S_2 = \int \frac{x^2 dx}{\sqrt{f(x)}},$$

$$S_{-1} = \int \frac{dx}{(x - \alpha)\sqrt{f(x)}},$$

where S_{-1} still depends on the parameter α .

II. If, however, $f(x)$ is of the third degree, the three integrals

$$S_0 = \int \frac{dx}{\sqrt{f(x)}}, \quad S_1 = \int \frac{x dx}{\sqrt{f(x)}}, \quad S_{-1} = \int \frac{dx}{(x - \alpha)\sqrt{f(x)}}$$

suffice.

In both cases S_0 is of the first kind. If $f(x)$ is of the third degree, then S_1 is of the second kind and S_{-1} of the third kind. If, on the other hand, $f(x)$ is of the fourth degree, then S_1, S_2 and S_{-1} are of the third kind. But from S_1 and S_2 one can form an integral of the second kind. For

$$\frac{1}{\sqrt{f(x)}} = (x^4 + a_1x^3 + a_2x^2 + a_3x + a_4)^{-1/2} = x^{-2} - \frac{a_1}{2}x^{-3} + \dots,$$

and in the development of

$$\left(x^2 + \frac{a_1}{2}x\right) : \sqrt{f(x)}$$

in descending powers no term with x^{-1} occurs. Consequently

$$S_2 + \frac{a_1}{2}S_1$$

is an integral of the second kind.

By the mediation of a logarithmic function one can also reduce the four integrals of case I to three. To this end bring $f(x)$ into the form

$$x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = P^2 + ax + b,$$

where

$$P = x^2 + px + q$$

is a quadratic function and a and b are constants. It follows that

$$p = \frac{a_1}{2}, \quad q = \frac{a_2}{2} - \frac{a_1^2}{8},$$

$$a = a_3 - \frac{a_1a_2}{2} + \frac{a_1^3}{8}, \quad b = a_4 - \left(\frac{a_2}{2} - \frac{a_1^2}{8}\right)^2.$$

Now

$$d \log \frac{\sqrt{f} - P}{\sqrt{f} + P} = \frac{f'(x)P - 2f(x)P'(x)}{f - P^2} \frac{dx}{\sqrt{f(x)}}, \quad (3)$$

$$f - P^2 = ax + b, \quad f'(x) = 2PP' + a.$$

and consequently

$$f'(x)P - 2f(x)P'(x) = a(P - 2xP') - 2bP' = Q,$$

a quadratic function of x , and therefore from (3)

$$\log \frac{\sqrt{f} - P}{\sqrt{f} + P} = \int \frac{Q}{ax + b} \frac{dx}{\sqrt{f(x)}}. \quad (4)$$

The right hand side, however, is a linear function of S_0, S_1 and S_{-1} (for $\alpha = -b/a$). In the special case where $a = 0$, Q is of the first degree, and one can express a linear combination of S_0 and S_{-1} by a logarithmic function.

If we assume a_1 and a_3 to be 0, to which the general case can be reduced by a linear transformation according to § 4, then we obtain as fundamental integrals

$$S_0 = \int \frac{dx}{\sqrt{f(x)}}, \quad S_2 = \int \frac{x^2 dx}{\sqrt{f(x)}}, \quad S_{-1} = \int \frac{dx}{(x - \alpha)\sqrt{f(x)}},$$

whereas $S_1 = \int \frac{x dx}{\sqrt{f(x)}}$ is reduced by the substitution $x^2 = y$ to the non-elliptic integral

$$\frac{1}{2} \int \frac{dy}{\sqrt{y^2 + a_2 y + a_4}}.$$

Here S_0 is of the first kind, S_2 of the second, S_{-1} of the third kind. For Legendre's normal form we put $f(x) = (1 - x^2)(1 - \kappa^2 x^2)$ and take as the normal integral of the second kind not S_2 itself, but $S_0 - \kappa^2 S_2$. In this way the normal integrals of the three kinds are

$$\int \frac{dx}{\sqrt{(1 - x^2)(1 - \kappa^2 x^2)}}, \quad \int \sqrt{\frac{1 - \kappa^2 x^2}{1 - x^2}} dx, \quad \int \frac{dx}{(x - \alpha)\sqrt{(1 - x^2)(1 - \kappa^2 x^2)}},$$

and, if one puts $x = \sin \varphi$,

$$\int \frac{d\varphi}{\sqrt{1 - \kappa^2 \sin^2 \varphi}}, \quad \int \sqrt{1 - \kappa^2 \sin^2 \varphi} d\varphi, \quad \int \frac{d\varphi}{(\sin \varphi - \alpha)\sqrt{1 - \kappa^2 \sin^2 \varphi}}.$$

The same result is obtained if one assumes the elliptic differential in the normal form

$$\frac{dz}{\sqrt{z(1 - z)(1 - \kappa^2 z)}}$$

and proceeds according to case II.

§ 13. The addition theorem.

The addition theorem of elliptic integrals, discovered by Euler, consists in the theorem, fundamental for the whole further theory, that if, of the three pairs of values

$$x_1, \sqrt{f(x_1)}; \quad x_2, \sqrt{f(x_2)}; \quad x_3, \sqrt{f(x_3)}$$

two are assumed given, one can determine the third algebraically in such a way that the differential equation

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} = 0 \quad (1)$$

is satisfied. Here $f(x)$ is a given function of the third or fourth degree. This is a special case of Abel's great theorem, and is also to be understood and derived here in that way¹. We precede the proof by the following elementary algebraic theorem.

Let $F(x)$ be an integral rational function of degree n without multiple factors, $F'(x)$ its derivative, let $x_1, x_2, \dots, x_n$ be the roots of the equation $F(x) = 0$, and let $\varphi(x)$ be an integral function of x whose degree is at most $n - 2$. Then

$$\frac{\varphi(x_1)}{F'(x_1)} + \frac{\varphi(x_2)}{F'(x_2)} + \dots + \frac{\varphi(x_n)}{F'(x_n)} = 0. \quad (2)$$

The proof of this theorem follows immediately from decomposing the rational fraction

$$\frac{x\varphi(x)}{F(x)}$$

into partial fractions

$$\frac{x_1\varphi(x_1)}{F'(x_1)(x - x_1)} + \frac{x_2\varphi(x_2)}{F'(x_2)(x - x_2)} + \dots + \frac{x_n\varphi(x_n)}{F'(x_n)(x - x_n)},$$

and then putting $x = 0$ [Vol. I, § 15 (9)].

Let now P, Q denote integral functions of x of degrees m and $m - 2$, and let us ask for the values of $x, \sqrt{f(x)}$ for which the function

$$P + Q\sqrt{f(x)} \quad (3)$$

vanishes. There are $2m$ such pairs of values (points), and the values of x are found as roots of the equation of degree $2m$:

$$F(x) = P^2 - Q^2f(x) = 0, \quad (4)$$

and the corresponding values of $\sqrt{f(x)}$ from

$$P + Q\sqrt{f(x)} = 0. \quad (5)$$

We now assume that the coefficients in the functions P, Q are variable, either independent variables or in some way dependent on other variables. Then the values x and $\sqrt{f(x)}$ determined by (4), (5) are also functions of these variables, and (5) can be differentiated.

Let δ denote differentiation with respect to the variables occurring in the coefficients of P, Q , where x is regarded as constant; and let dx denote the corresponding differential of x when x is determined by (4) or (5) as a function of these coefficients. Differentiating (4) gives

$$2P\delta P - 2Q\delta Qf(x) + F'(x)dx = 0,$$

whence, using (5),

$$\frac{Q\delta P - P\delta Q}{F'(x)} = \frac{dx}{2\sqrt{f(x)}}. \quad (6)$$

Since the numerator on the left has degree $2m - 2$, while $F'(x)$ has degree $2m$, formula (2) can be applied, and it follows that

$$\sum \frac{dx}{\sqrt{f(x)}} = 0, \quad (7)$$

¹This very comprehensive theorem is derived with magnificent simplicity in a scarcely two-page paper in the fourth volume of Crelle's Journal: "Démonstration d'une propriété générale d'une certaine classe de fonctions transcendentes"; Oeuvres complètes de N. H. Abel. Nouvelle édition, T. I, p. 515. Detailed presentation and applications are found in the posthumous large memoir: "Mémoire sur une propriété générale d'une classe très étendue de fonctions transcendentes".

where the sum is extended over all roots of the equation (5).

We apply this theorem to the simplest case, namely $m = 2$, and obtain the following theorem.

If

$$x_1, \sqrt{f(x_1)}; \quad x_2, \sqrt{f(x_2)}; \quad x_3, \sqrt{f(x_3)}; \quad x_4, \sqrt{f(x_4)} \quad (8)$$

are four pairs of values for which some function of the form

$$a + bx + cx^2 + \sqrt{f(x)} \quad (9)$$

vanishes, then these pairs of values are not wholly independent of one another, but between them there exists the differential equation

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} + \frac{dx_4}{\sqrt{f(x_4)}} = 0. \quad (10)$$

The dependence of these four pairs of values, that is, an integration of the differential equation (10), can also be expressed algebraically by putting the function (9) equal to zero for $x = x_1, x_2, x_3, x_4$ and then eliminating a, b, c . One obtains the determinant equation

$$\begin{vmatrix} 1 & x_1 & x_1^2 & \sqrt{f(x_1)} \\ 1 & x_2 & x_2^2 & \sqrt{f(x_2)} \\ 1 & x_3 & x_3^2 & \sqrt{f(x_3)} \\ 1 & x_4 & x_4^2 & \sqrt{f(x_4)} \end{vmatrix} = 0. \quad (11)$$

From this equation x_1 and $\sqrt{f(x_1)}$ can be expressed rationally through $x_2, \sqrt{f(x_2)}; x_3, \sqrt{f(x_3)}; x_4, \sqrt{f(x_4)}$; for x_1 is the root of a biquadratic equation whose three other roots are x_2, x_3, x_4 , and whose coefficients depend rationally on $x_2, \sqrt{f(x_2)}; x_3, \sqrt{f(x_3)}; x_4, \sqrt{f(x_4)}$, while $\sqrt{f(x_1)}$ is representable rationally through x_1 by means of (11). If x_4 is set equal to an arbitrary constant, then the differential equation (10) passes into (1), and (11) gives its integral.

We first apply this theorem to the normal form and put

$$f(x) = x(1-x)(1-\kappa^2 x).$$

If in (11) we assume $x_4 = 0$, then this equation reduces, after division by $\sqrt{x_1 x_2 x_3}$, to

$$\begin{vmatrix} \sqrt{x_1} & x_1 \sqrt{x_1} & \sqrt{(1-x_1)(1-\kappa^2 x_1)} \\ \sqrt{x_2} & x_2 \sqrt{x_2} & \sqrt{(1-x_2)(1-\kappa^2 x_2)} \\ \sqrt{x_3} & x_3 \sqrt{x_3} & \sqrt{(1-x_3)(1-\kappa^2 x_3)} \end{vmatrix} = 0, \quad (12)$$

or, expanding the determinant by the elements of the first row,

$$\sqrt{x_1}(a + bx_1) + c\sqrt{(1-x_1)(1-\kappa^2 x_1)} = 0, \quad (13)$$

where, for abbreviation,

$$\begin{aligned} a &= x_2 \sqrt{x_2} \sqrt{(1-x_3)(1-\kappa^2 x_3)} - x_3 \sqrt{x_3} \sqrt{(1-x_2)(1-\kappa^2 x_2)}, \\ b &= -\sqrt{x_2} \sqrt{(1-x_3)(1-\kappa^2 x_3)} + \sqrt{x_3} \sqrt{(1-x_2)(1-\kappa^2 x_2)}, \\ c &= -(x_2 - x_3) \sqrt{x_2} \sqrt{x_3}. \end{aligned} \quad (14)$$

If we rationalize (13), we obtain the cubic equation

$$x(a + bx)^2 - c^2(1-x)(1-\kappa^2 x) = 0,$$

whose three roots are x_1, x_2, x_3 . It is therefore identically

$$b^2(x - x_1)(x - x_2)(x - x_3) = x(a + bx)^2 - c^2(1 - x)(1 - \varkappa^2 x). \quad (15)$$

Putting $x = 0, 1, 1 : \varkappa^2$ in this identity, one obtains

$$\begin{aligned} \sqrt{x_1 x_2 x_3} &= -\frac{c}{b}, \\ \sqrt{(1 - x_1)(1 - x_2)(1 - x_3)} &= \frac{b + a}{b}, \\ \sqrt{(1 - \varkappa^2 x_1)(1 - \varkappa^2 x_2)(1 - \varkappa^2 x_3)} &= \frac{b + \varkappa^2 a}{b}. \end{aligned} \quad (16)$$

To determine the signs in these equations we further obtain from (13), by replacing x_1 by x_2 and x_3 and multiplying the results,

$$x_1 x_2 x_3 (a + bx_1)(a + bx_2)(a + bx_3) = -c^3 \sqrt{f(x_1)} \sqrt{f(x_2)} \sqrt{f(x_3)},$$

and (15) gives for $x = -a : b$

$$(a + bx_1)(a + bx_2)(a + bx_3) = \frac{c^2}{b^2} (b + a)(b + \varkappa^2 a).$$

If one also uses the square of the first equation (16), it follows that

$$\sqrt{f(x_1)} \sqrt{f(x_2)} \sqrt{f(x_3)} = -\frac{c}{b^3} (b + a)(b + \varkappa^2 a).$$

The same result, however, is obtained by multiplying the three equations (16), and thus the signs in these formulae are chosen so that the product of the left hand sides gives the value of $\sqrt{f(x_1)}$ required by the differential equation (1). A further determination of the signs in (16) is not possible by the nature of the matter, and these equations serve for the unambiguous definition of $\sqrt{x_1}$, $\sqrt{1 - x_1}$, $\sqrt{1 - \varkappa^2 x_1}$ when $\sqrt{x_2}$, $\sqrt{1 - x_2}$, $\sqrt{1 - \varkappa^2 x_2}$, $\sqrt{x_3}$, $\sqrt{1 - x_3}$, $\sqrt{1 - \varkappa^2 x_3}$ are assumed given.

To bring equations (16) into the customary form, we first rationalize the numerator of b in (14) and obtain

$$b = -\frac{(x_2 - x_3)(1 - \varkappa^2 x_2 x_3)}{\sqrt{x_2} \sqrt{(1 - x_3)(1 - \varkappa^2 x_3)} + \sqrt{x_3} \sqrt{(1 - x_2)(1 - \varkappa^2 x_2)}},$$

and further, by (14),

$$\begin{aligned} \frac{b + a}{b} &= \frac{\sqrt{1 - x_2} \sqrt{1 - x_3} - \sqrt{x_2 x_3} \sqrt{(1 - \varkappa^2 x_2)(1 - \varkappa^2 x_3)}}{1 - \varkappa^2 x_2 x_3}, \\ \frac{b + \varkappa^2 a}{b} &= \frac{\sqrt{1 - \varkappa^2 x_2} \sqrt{1 - \varkappa^2 x_3} - \varkappa^2 \sqrt{x_2 x_3} \sqrt{(1 - x_2)(1 - x_3)}}{1 - \varkappa^2 x_2 x_3}. \end{aligned}$$

Thus equations (16) are easily brought into the form

$$\begin{aligned} \sqrt{x_1} &= -\frac{\sqrt{x_2} \sqrt{(1 - x_3)(1 - \varkappa^2 x_3)} + \sqrt{x_3} \sqrt{(1 - x_2)(1 - \varkappa^2 x_2)}}{1 - \varkappa^2 x_2 x_3}, \\ \sqrt{1 - x_1} &= \frac{\sqrt{1 - x_2} \sqrt{1 - x_3} - \sqrt{x_2 x_3} \sqrt{(1 - \varkappa^2 x_2)(1 - \varkappa^2 x_3)}}{1 - \varkappa^2 x_2 x_3}, \\ \sqrt{1 - \varkappa^2 x_1} &= \frac{\sqrt{1 - \varkappa^2 x_2} \sqrt{1 - \varkappa^2 x_3} - \varkappa^2 \sqrt{x_2 x_3} \sqrt{(1 - x_2)(1 - x_3)}}{1 - \varkappa^2 x_2 x_3}. \end{aligned} \quad (17)$$

and our consideration shows that, when x_1 and the roots $\sqrt{x_1}$, $\sqrt{1-x_1}$, $\sqrt{1-\kappa^2 x_1}$ are determined by these equations as functions of x_2, x_3 , the differential equation (1) is identically satisfied.

In order also to obtain the addition theorem for Weierstrass's normal form in the simplest possible shape, put

$$f(x) = 4x^3 - g_2x - g_3,$$

and in equation (11), after division by x_4^2 , let x_4 become infinitely large. Then the integral of the differential equation (1)

$$\frac{dx_1}{\sqrt{f(x_1)}} + \frac{dx_2}{\sqrt{f(x_2)}} + \frac{dx_3}{\sqrt{f(x_3)}} = 0$$

is obtained in the form

$$\begin{vmatrix} 1 & x_1 & \sqrt{f(x_1)} \\ 1 & x_2 & \sqrt{f(x_2)} \\ 1 & x_3 & \sqrt{f(x_3)} \end{vmatrix} = 0, \quad (18)$$

or

$$a + bx_1 + c\sqrt{f(x_1)} = 0, \quad (19)$$

where

$$a = x_2\sqrt{f(x_3)} - x_3\sqrt{f(x_2)}, \quad b = \sqrt{f(x_2)} - \sqrt{f(x_3)}, \quad c = x_3 - x_2. \quad (20)$$

Then x_1, x_2, x_3 become the roots of the cubic equation

$$(a + bx)^2 - c^2 f(x) = 0,$$

and, if the coefficient of x^2 divided by that of x^3 is set equal to the negative sum of the roots, one obtains

$$x_1 + x_2 + x_3 = \frac{1}{4} \left(\frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3} \right)^2, \quad (21)$$

and from (19)

$$\sqrt{f(x_1)} = \frac{1}{4} \left(\frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3} \right)^3 - (x_2 + x_3) \frac{\sqrt{f(x_2)} - \sqrt{f(x_3)}}{x_2 - x_3} - \frac{x_3\sqrt{f(x_2)} - x_2\sqrt{f(x_3)}}{x_2 - x_3}. \quad (22)$$

§ 14. Origin of the elliptic functions.

If in the elliptic differential of the first kind in Legendre's normal form

$$\frac{1}{2} \frac{dz}{\sqrt{z(1-z)(1-\kappa^2 z)}}$$

the substitution

$$z = \sin^2 \varphi \quad (1)$$

is made, and the integration is then carried out from $\varphi = 0$, then there arises the elliptic integral of the first kind

$$\int_0^\varphi \frac{d\varphi}{\sqrt{1 - \kappa^2 \sin^2 \varphi}} = u. \quad (2)$$

Jacobi calls the upper limit φ of this integral its amplitude and writes

$$\varphi = \operatorname{am} u. \quad (3)$$

The trigonometric functions of this arc,

$$\sin \varphi, \quad \cos \varphi, \quad \sqrt{1 - \kappa^2 \sin^2 \varphi} = \Delta \varphi, \quad (4)$$

are now precisely those which, considered as functions of u , are called elliptic functions and denoted by Jacobi by

$$\sin \operatorname{am} u, \quad \cos \operatorname{am} u, \quad \Delta \operatorname{am} u,$$

or, after Gudermann, more briefly by

$$\operatorname{sn} u, \quad \operatorname{cn} u, \quad \operatorname{dn} u. \quad (5)$$

These functions depend on the two arguments u and κ^2 , and if a more exact notation is necessary one also writes

$$\operatorname{sn}(u, \kappa^2), \quad \operatorname{cn}(u, \kappa^2), \quad \operatorname{dn}(u, \kappa^2).$$

If in formula (1) φ is transformed into $-\varphi$, then u passes into $-u$; hence $\operatorname{am}(-u) = -\operatorname{am} u$, and consequently

$$\operatorname{sn}(-u) = -\operatorname{sn} u, \quad \operatorname{cn}(-u) = \operatorname{cn} u, \quad \operatorname{dn}(-u) = \operatorname{dn} u,$$

that is, $\operatorname{sn} u$ is an odd function, while $\operatorname{cn} u$ and $\operatorname{dn} u$ are even functions.

If therefore one puts

$$\frac{d\varphi_2}{\Delta\varphi_2} = du, \quad \frac{d\varphi_3}{\Delta\varphi_3} = dv, \quad \frac{d\varphi_1}{\Delta\varphi_1} = -d(u+v),$$

then the formulae (17), § 13, can be applied, and one obtains the addition theorems:

$$\begin{aligned} \operatorname{sn}(u+v) &= \frac{\operatorname{sn} u \operatorname{cn} v \operatorname{dn} v + \operatorname{sn} v \operatorname{cn} u \operatorname{dn} u}{1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \\ \operatorname{cn}(u+v) &= \frac{\operatorname{cn} u \operatorname{cn} v - \operatorname{sn} u \operatorname{sn} v \operatorname{dn} u \operatorname{dn} v}{1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \\ \operatorname{dn}(u+v) &= \frac{\operatorname{dn} u \operatorname{dn} v - \kappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{cn} u \operatorname{cn} v}{1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}. \end{aligned} \quad (6)$$

From these one can compute the three elliptic functions unambiguously for arbitrary values of the argument, once they are determined for any, however small, region around the origin. Weierstrass takes this theorem as the point of departure for the theory of elliptic functions.

If one puts

$$\int_0^{\pi/2} \frac{d\varphi}{\Delta\varphi} = K,$$

then $\operatorname{sn} K = 1$, $\operatorname{cn} K = 0$, $\operatorname{dn} K = \sqrt{1 - \kappa^2} = \kappa'$, and formulae (6), with $v = K$, give

$$\operatorname{sn}(u+K) = \frac{\operatorname{cn} u}{\operatorname{dn} u}, \quad \operatorname{cn}(u+K) = \frac{-\kappa' \operatorname{sn} u}{\operatorname{dn} u}, \quad \operatorname{dn}(u+K) = \frac{\kappa'}{\operatorname{dn} u}, \quad (7)$$

and if this formula is applied once more:

$$\operatorname{sn}(u+2K) = -\operatorname{sn} u, \quad \operatorname{cn}(u+2K) = -\operatorname{cn} u, \quad \operatorname{dn}(u+2K) = \operatorname{dn} u, \quad (8)$$

and applied again:

$$\operatorname{sn}(u+4K) = \operatorname{sn} u, \quad \operatorname{cn}(u+4K) = \operatorname{cn} u, \quad \operatorname{dn}(u+4K) = \operatorname{dn} u. \quad (9)$$

Thus the elliptic functions have the property of periodicity in common with the trigonometric functions. They have the period $4K$ ($\operatorname{dn} u$ also has the period $2K$).

In § 3, as an example, we derived the transformation, formula (1):

$$\frac{dz}{\sqrt{z(1-z)(1-\kappa^2 z)}} = \frac{dx}{i\sqrt{x(1-x)(1-\kappa'^2 x)}}, \quad (10)$$

when

$$z = \frac{-x}{1-x} \quad (11)$$

held. Therefore, if one puts

$$\frac{dx}{\sqrt{x(1-x)(1-\kappa'^2 x)}} = i du,$$

then, if x and u vanish simultaneously, $x = \operatorname{sn}^2(iu, \kappa')$, $1-x = \operatorname{cn}^2(iu, \kappa')$, and from (10) it follows that

$$\frac{dz}{\sqrt{z(1-z)(1-\kappa^2 z)}} = du,$$

hence

$$z = \operatorname{sn}^2(u, \kappa).$$

Accordingly (11) gives

$$\operatorname{sn}(u, \kappa) = \frac{-i \operatorname{sn}(iu, \kappa')}{\operatorname{cn}(iu, \kappa')} \quad (12)$$

(where the sign is to be determined from the special value $u = 0$).

If now

$$\int_0^{\pi/2} \frac{d\varphi}{\sqrt{1-\kappa'^2 \sin^2 \varphi}} = K'$$

is put, then from (8) one obtains

$$\operatorname{sn}(iu + 2K', \kappa') = -\operatorname{sn}(iu, \kappa'), \quad \operatorname{cn}(iu + 2K', \kappa') = -\operatorname{cn}(iu, \kappa'),$$

and consequently from (12)

$$\operatorname{sn}(u - 2iK', \kappa) = \operatorname{sn}(u, \kappa),$$

or also, by replacing u by $u + 2iK'$ and omitting the notation κ again:

$$\operatorname{sn}(u + 2iK') = \operatorname{sn} u. \quad (13)$$

Thus the function $\operatorname{sn} u$ has a double periodicity. One of these periods, $4K$, is real, the other, $2iK'$, purely imaginary (at least when the modulus κ is a positive proper fraction). This double periodicity is a property which occurs nowhere in the realm of the elementary functions, and thus showed most clearly that one was dealing here with a new kind of function. At the same time, the consequences derived from double periodicity are those which, as experience has shown, lead by far the most quickly and with satisfactory rigor into the interior of the theory, so that here the most suitable point of departure for a methodical development is to be sought. Our next considerations will therefore be devoted to the doubly periodic functions in general.